ASPECT WAVE LLC

Vessel Measurement Recorder

November 2026 · Prepared by Sophie Scopazzi · Firmware v8.12

Wi-Fi Dashboard & OLED Screens

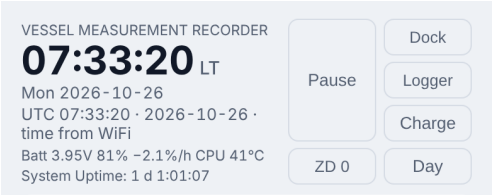
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Wi-Fi Dashboard

There dashboard is broken into cards. Every card can be hidden / shown as desired using a permanent card at the bottom of the dashboard. The information at the top outside of the cards cannot be hidden

Top Information and Buttons



The local time using the ZD retrieved from the connected device or manually set by tapping the button. Buttons for dock, logger, and charge mode and to change the display color (day, night, red)

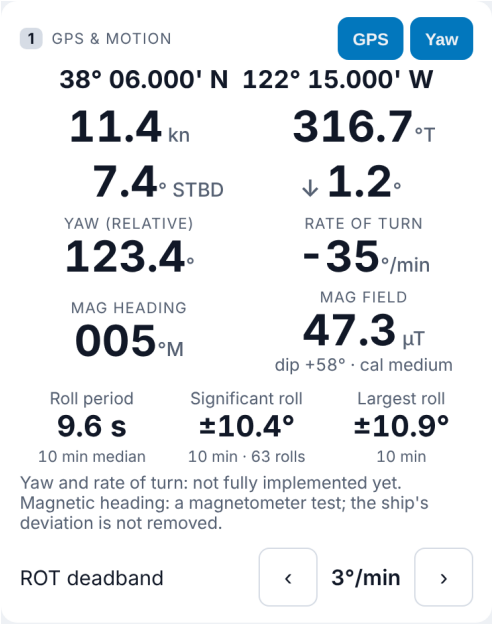
Dock Mode shows a hidden trim and list card at the top (viewing this card is only accessible via the button)

Logger Mode disables everything not required to save data to the microSD card (Wi-Fi, with longer sample-durations the GPS, and more)

Charge Mode turns everything off for fastest charging

Pause Button pauses the current recording, in case the VMR needs to be moved while recording. It does not pause the live data feed shown on the dashboard

Card 1 – GPS & Motion



Show and hide the GPS and Yaw / Magnetic data with buttons top right

With Yaw shown, the card also shows the magnetic heading and the magnetic field from the BNO085's magnetometer, which are logged in the CSV. The ship's own deviation is not removed

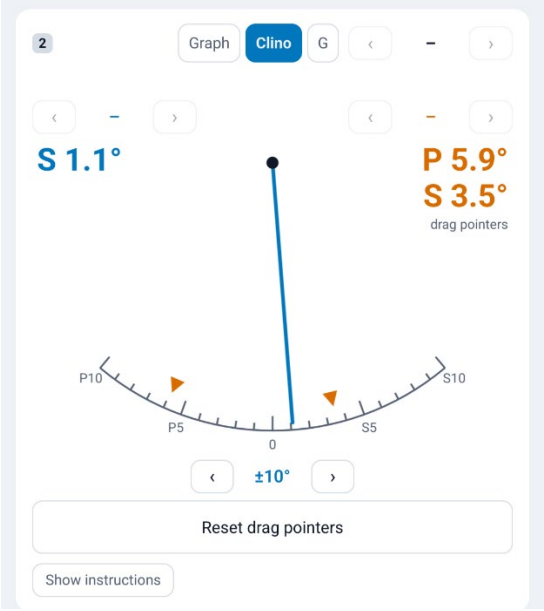
Variation is shown and input with a button between the magnetic heading and magnetic field (not yet in photo to left)

Card 2 – Live Data Graphs

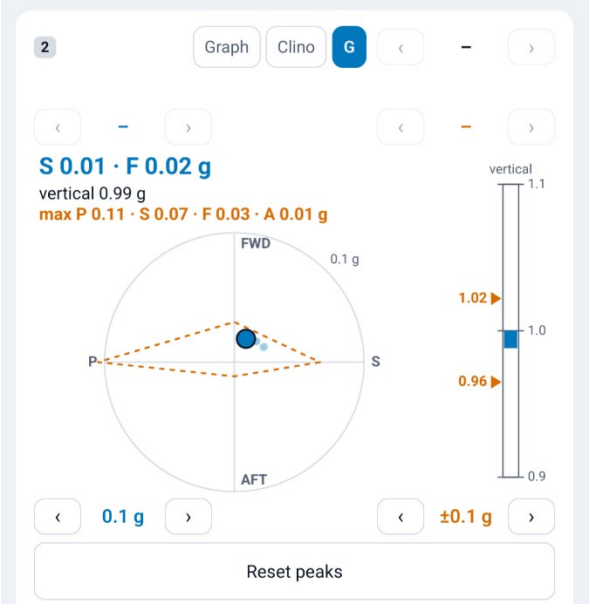


Graph

The OLED graph, clinometer, and g-meter mirrors the dashboard settings

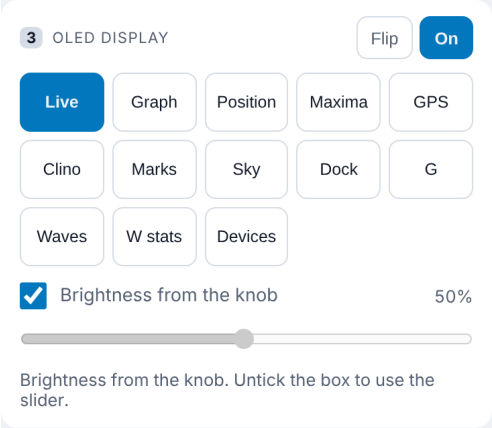


Clinometer



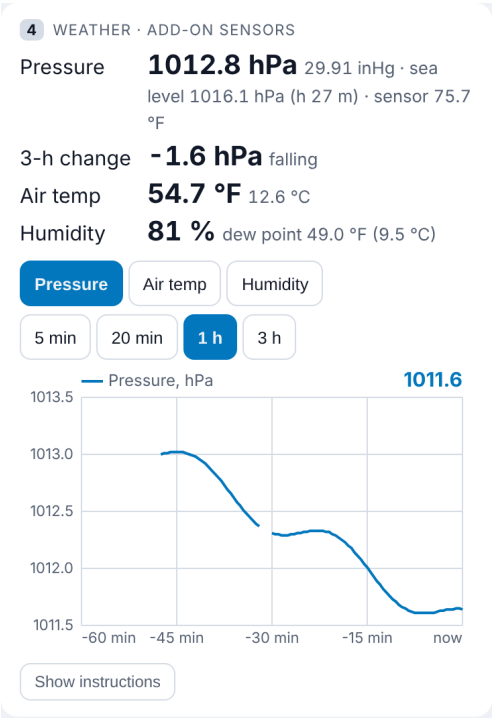
G-meter

Card 3 – OLED Display



Power on and off and 180° flip the OLED. Select what view to see, with brightness controlled from the Wi-Fi dashboard or from the knob

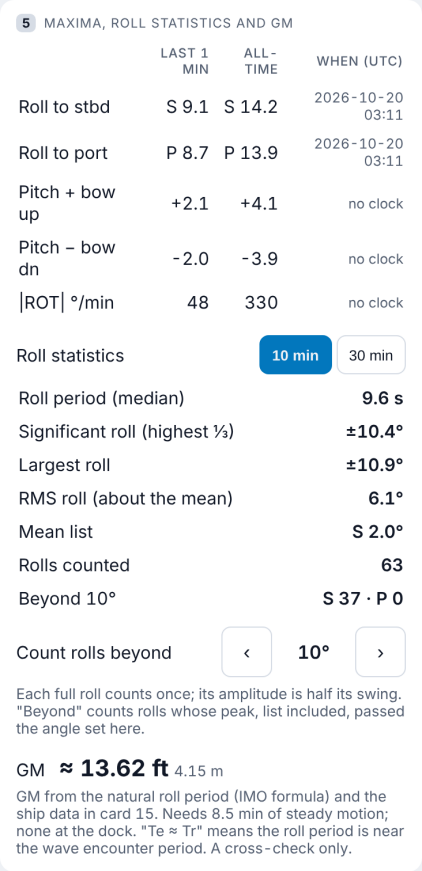
Card 4 – Weather



Pressure (also in inHg and reduced to sea level, if the vessel particulars, drafts, and VMR location onboard are entered), the 3-hour change, air temperature, humidity and dew point from the BMP388 and SHT30. Shown only when one of those sensors is switched on in card 17.

Tap Pressure, Air temp or Humidity for a live graph over 5 or 20 minutes, 1 or 3 hours. The dew point is drawn dashed with the air temperature: where they meet, expect fog or condensation.

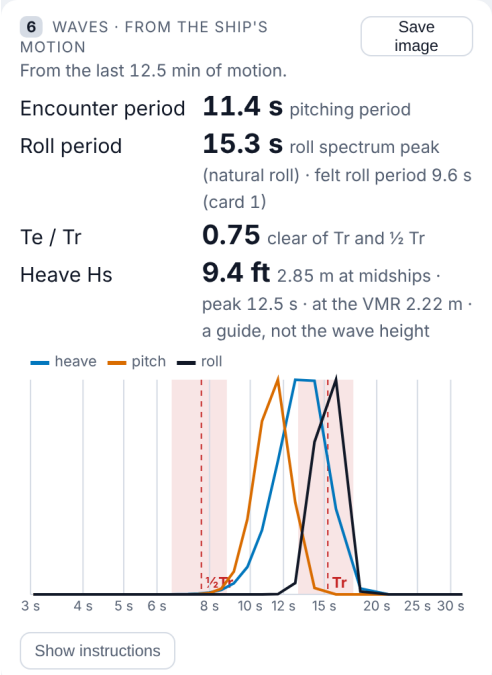
Card 5 – Maxima, Roll Statistics and GM



Statistics about roll and pitch with ability to count the number of rolls beyond a certain number. Shows the estimated GM. Requires the vessel particulars to calculate

Estimated GM uses

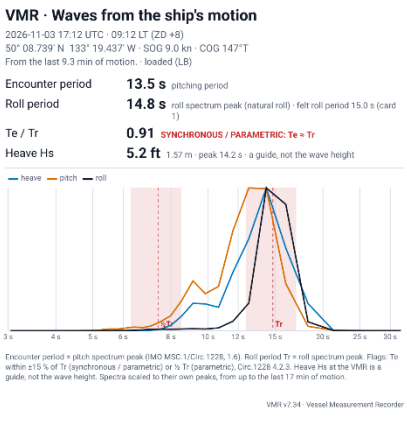
Card 6 – Waves



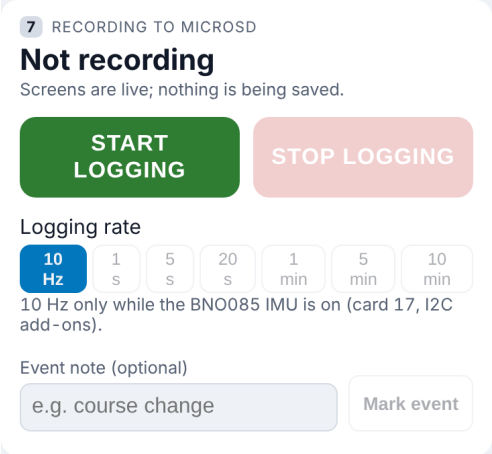
Waves graph detailing the encounter period, roll period, Te/Tr (metric for synchronous and parametric rolling) with an estimate of wave height from the heave (Hs)

The wave graph can be saved using the button on the top right

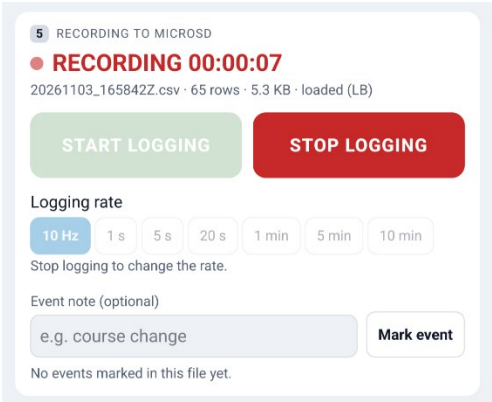
With where the VMR is entered in card 15, Hs is the heave at midships; without it, the heave at the VMR.



Card 7 – Recording to MicroSD



Not recording



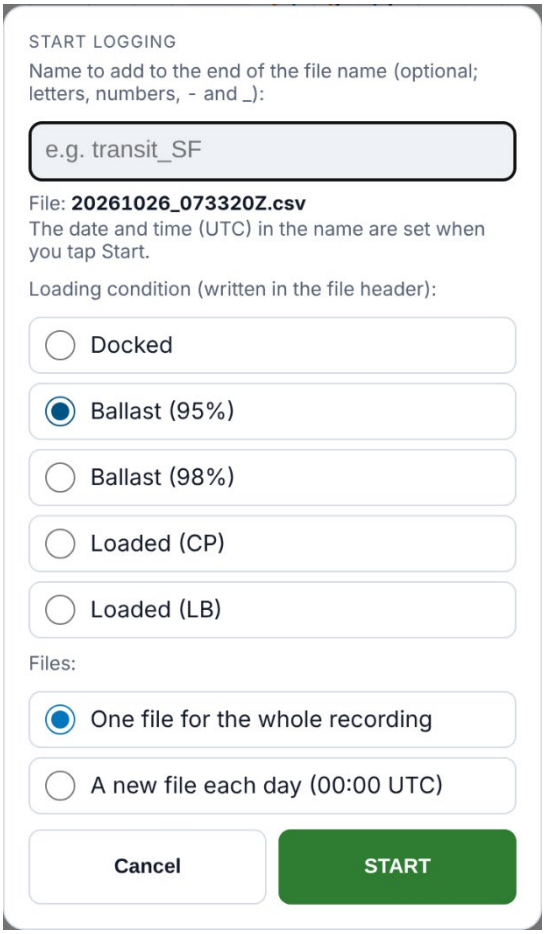
Recording

Start and stop logging. Change the logging frequency. With the IMU enabled the logging rate cannot be less than 10 Hz (other settings grayed out)

When the VMR is continuously logging, an event note can be added in this card that is saved in the CSV

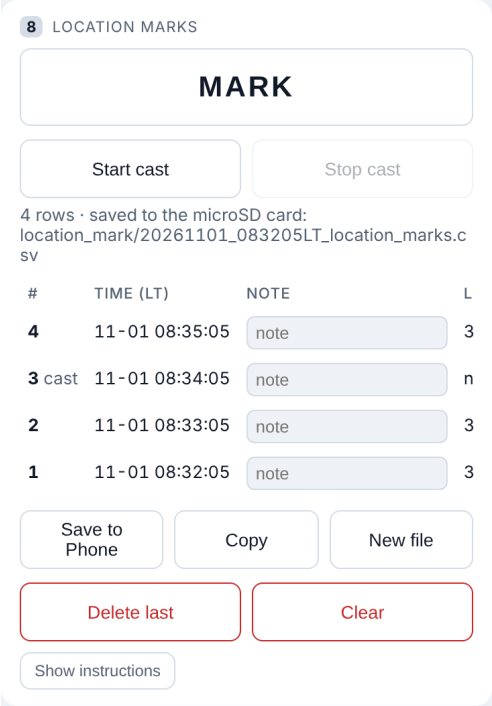
After hitting the green button, a popup appears where an addition can be made to the file name with settings for the ship's load condition (saved in the header) and whether the user desires one file or a new one to be created on each new UTC day.

Once recording, the card says the file name, how many rows it contains, and its size



Start Logging pop-up

Card 8 – Location Marks

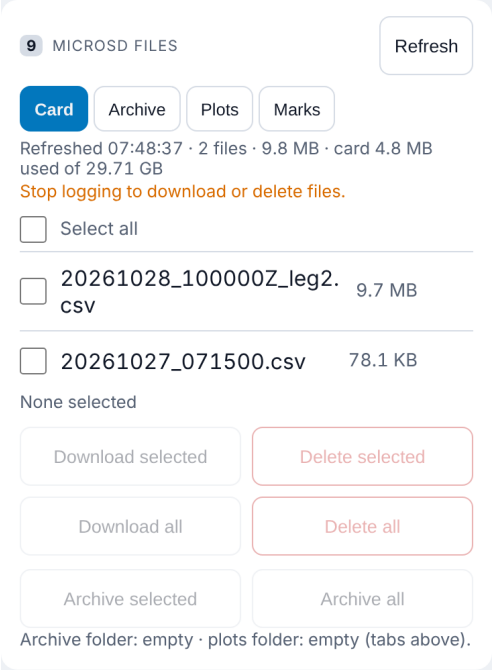


MARK records the time, position, COG / SOG and weather. Start cast and Stop cast record a cast as one row, with its duration and drift are automatically calculated. The physical Mark and Cast buttons on the VMR do the same.

A note can be typed on any row. Save to Phone downloads the marks CSV, Copy puts it on the clipboard, New file starts a new list and keeps the old file, and Clear deletes both. Delete last removes the newest row.

Opened with the Show Location Marks button.

Card 9 – MicroSD Files



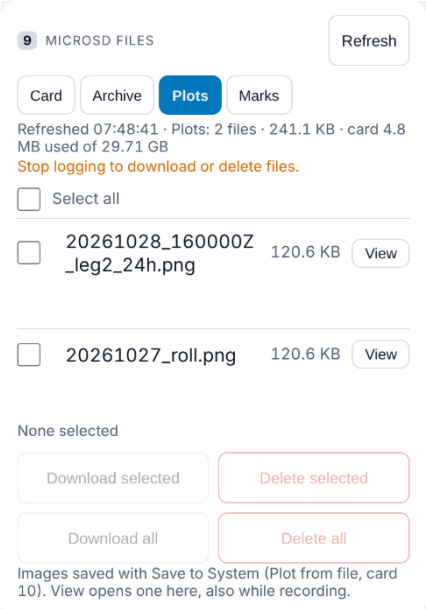
Main card

Viewer for the files on the microSD card. Files can be selected, downloaded, or archived

There is the main card, followed by folders for the archive, plots, and mark files

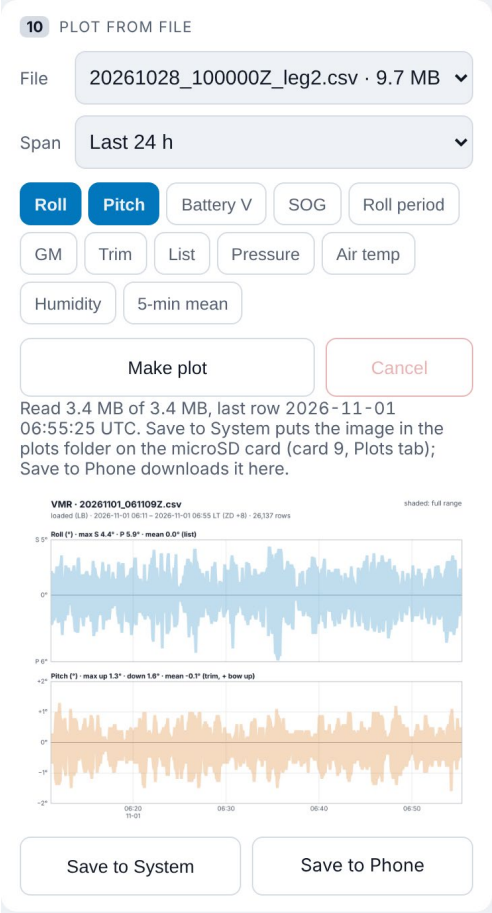
When the VMR is logging, files cannot be downloaded or deleted

On the plots page, images can be viewed by tapping the View button



Plots folder

Card 10 – Plot From File



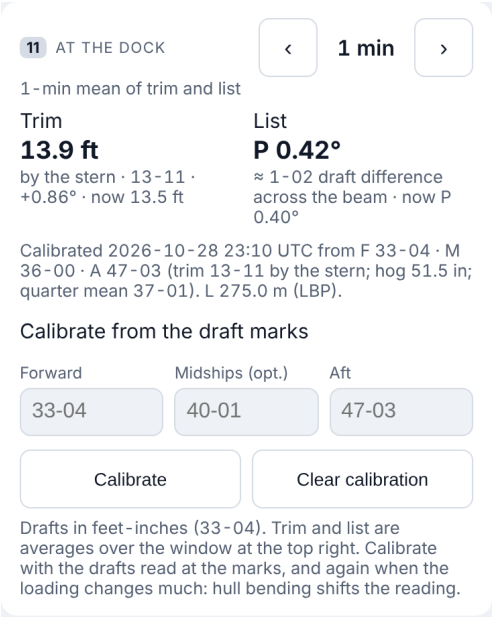
Generate plots from CSV files on the main Card (files not archived)

Select a duration of the file to plot – last 24 hrs, the full file, etc

Once a plot is generated, the user can select the desired variables which are automatically added / removed from the plot in the viewer

Save to System saves the plot to the VMR's Plots folder. Save to Phone saves the plot with the proper file name to the connected device

Card 11 – At the Dock



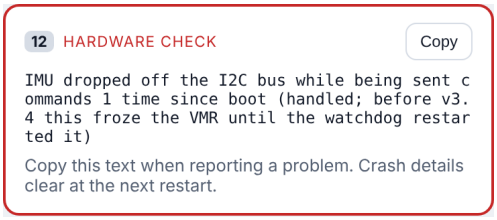
Shown at the top, in place of card 1, while Dock Mode is on (the Dock button). Trim in feet and list in degrees, averaged over 30 s to 5 min (set at the top right).

Calibrate by entering the forward, midships and aft drafts read at the marks. Recalibrate when the loading changes much (hull bending changes the deck slope near the ends). The wave analysis is off while at the dock.

Developer Settings

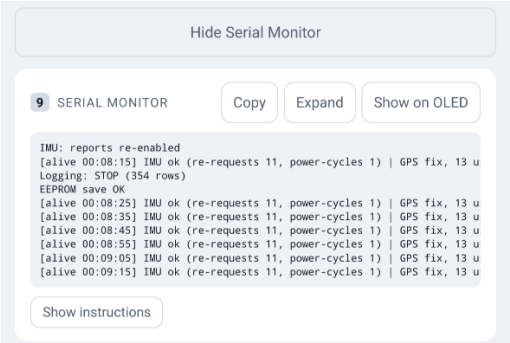
Cards beyond number eleven are hidden behind a Developer Settings button

Card 12 – Hardware Check



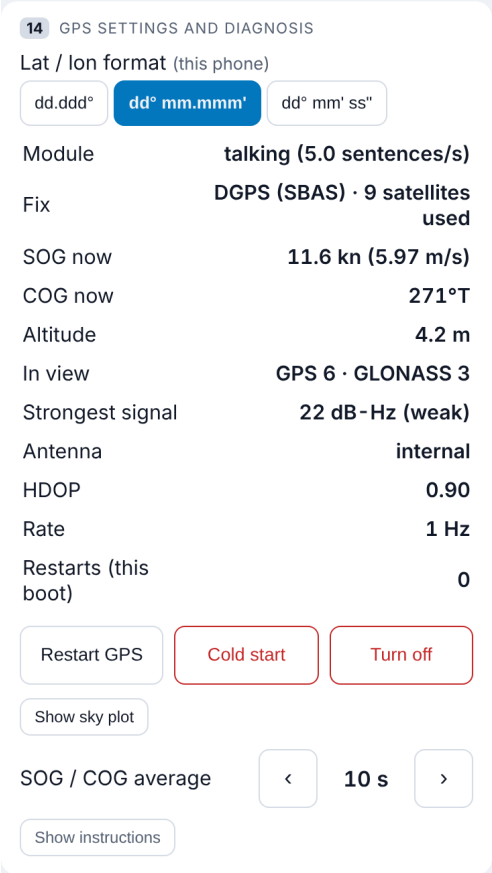
Displays errors with the system. The IMU dropping off the I2C bus is a fairly standard error – the VMR has routines to auto-retry connection

Card 13 – Serial Monitor



By default hidden; only requests data once the card is opened to save bandwidth. Expand makes the card 3x taller, scrolls right with no text wrapping, with Copy button to copy all that is displayed to the connected device's clipboard

Card 14 – GPS Settings and Diagnosis

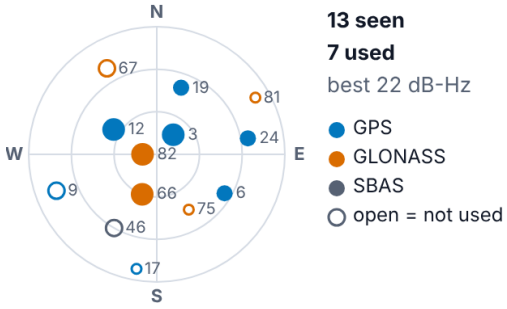


Select the GPS data format that is used to display throughout the VMR's screens. Default format is shown left in image

Health data from the GPS, whether it's talking, etc

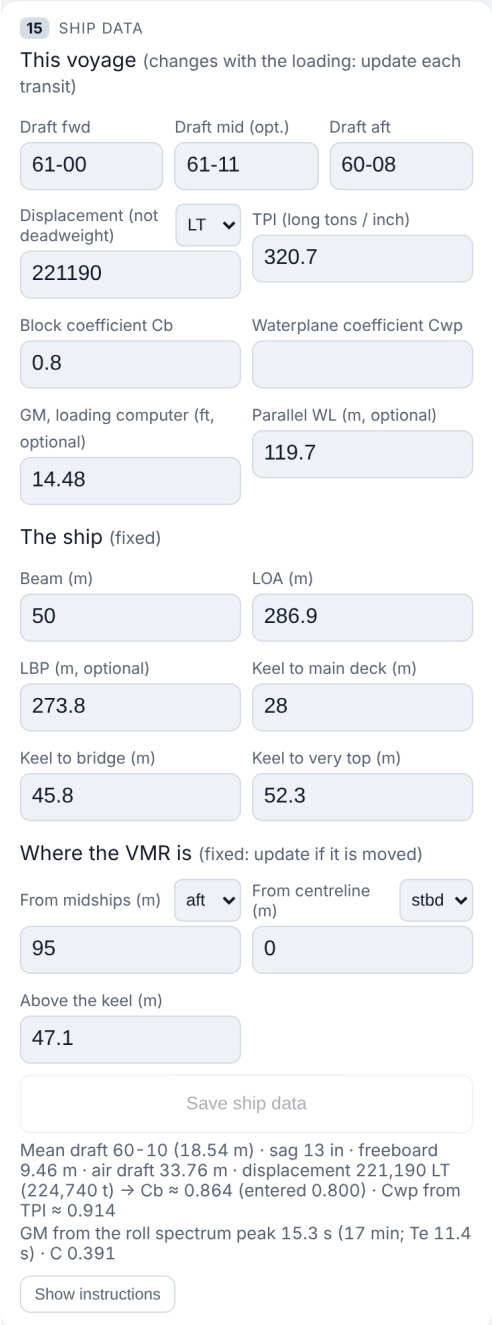
Buttons to restart, cold start, or turn off the GPS

At the bottom, the time to average the SOG / COG over. It defaults to 10 seconds



Satellites the GPS can see: horizon at the edge, overhead in the middle, north up. Filled = used in the fix; bigger = stronger signal.

Card 15 – Ship Data



Per-voyage data at the top and ship-specific data at the bottom (static). All saved to EEPROM to survive system restarts / firmware reflashes

Some values are optional. Without non-optional values input, GM will not be calculated

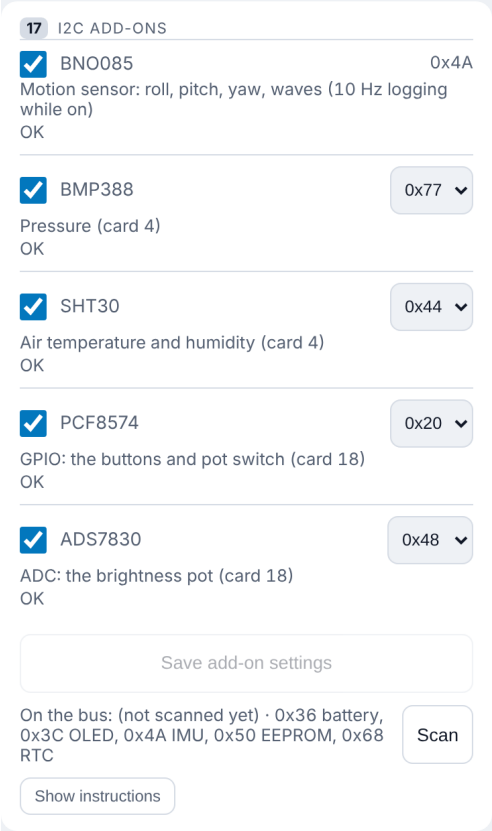
Where the VMR is: its distance from midships (fore or aft), from the centerline, and its height above the keel. This is used to move the heave to midships (card 6) to better estimate wave height and

Card 16 – Level Zero



Used to level the VMR when the vessel is in a static condition. Set zero button allows automatic setting with buttons for 0.1° changes

Card 17 – I2C Add-Ons

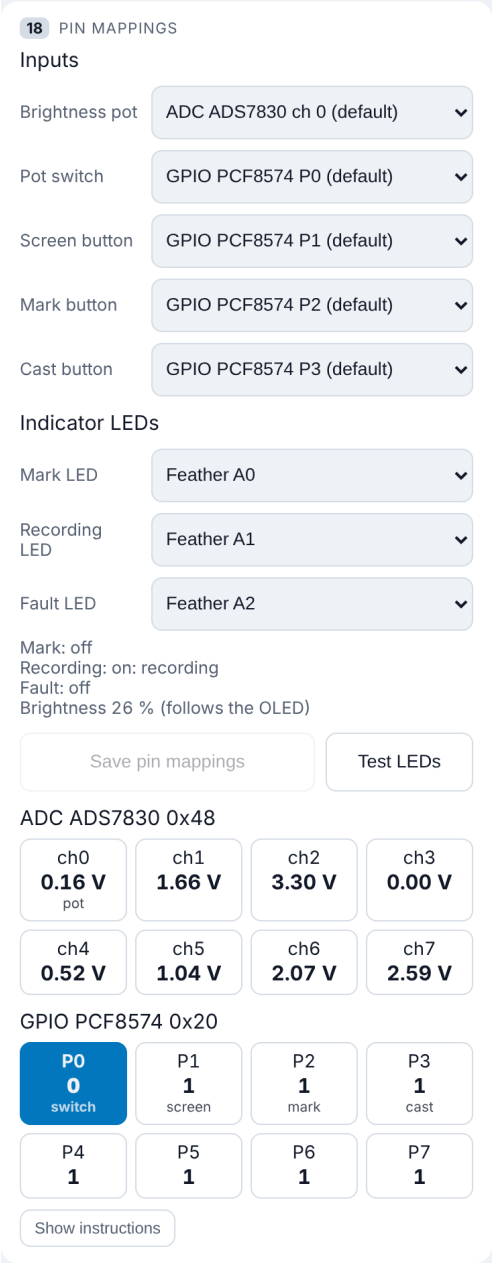


Select which I2C devices are in the system, or enable and disable boards

If an IMU is in the system, but the user doesn't want IMU data and to log slower than 10 Hz the IMU can be disabled thus allowing longer datalogging rates

Allows any I2C sensor to be added

Card 18 – Pin Mappings



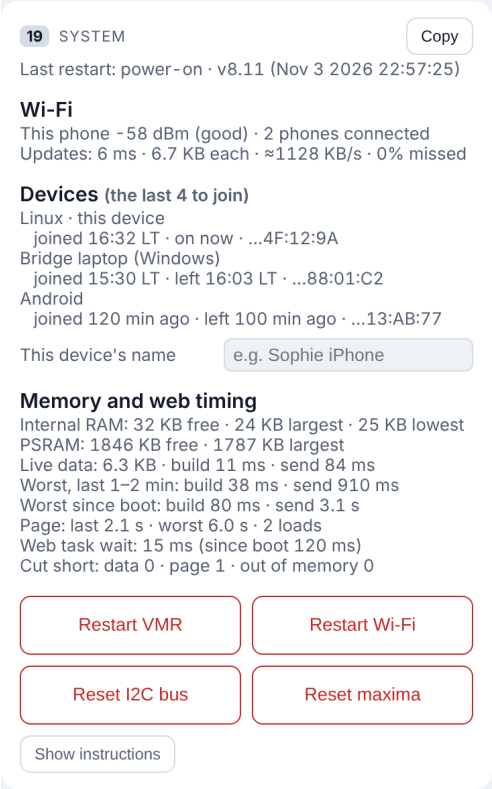
Which pins the controls and indicator LEDs use: the brightness pot (ADC), the pot switch, screen, Mark and Cast buttons (GPIO), and the Mark, Recording and Fault LEDs (Feather pins A0–A3)

Shows what each LED is showing and why the Fault LED is on, with the ADC channels and GPIO pins live. Test LEDs lights each LED in turn for 1 s

If built as default, no change in settings is required

Allows easy expandability for other sensors, controls, or indicators

Card 19 – System

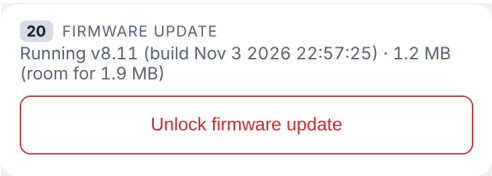


System-level information about the connected device and other connected devices

Memory diagnostics for help with troubleshooting

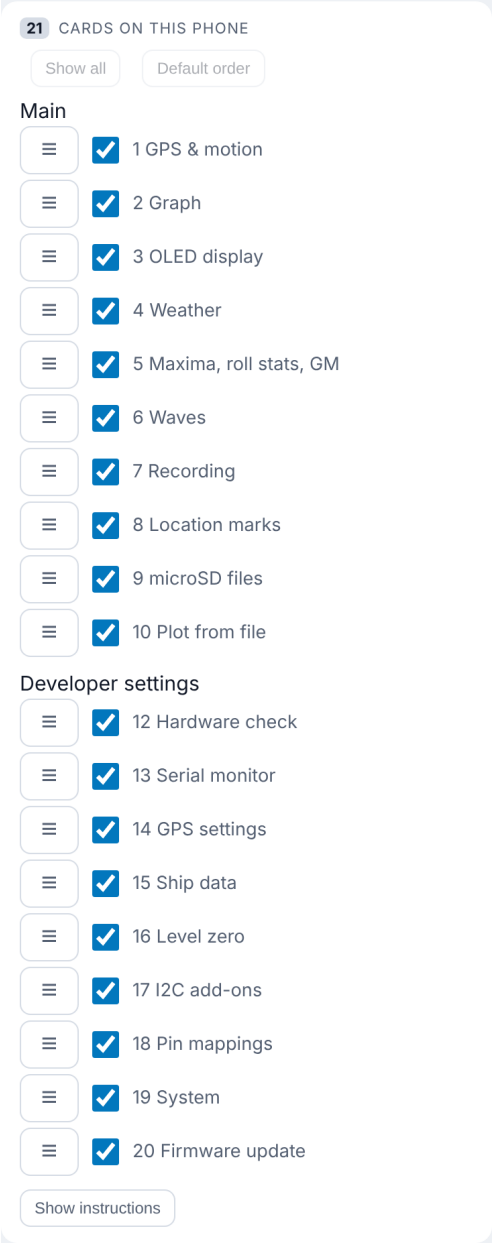
Buttons to control the systems

Card 20 – Firmware Update



Upload compiled .bin files to reflash the firmware over the air via Wi-Fi. Unlock firmware update asks for the firmware password first.

Card 21 – Cards on This Phone

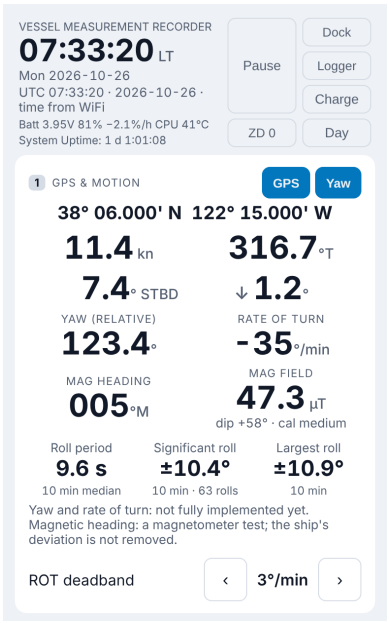


Untick a card to hide it on the connected device. Drag it by its ≡ handle to change the order. Each connected device keeps its own choice (does not change the order of cards for any other connected devices)

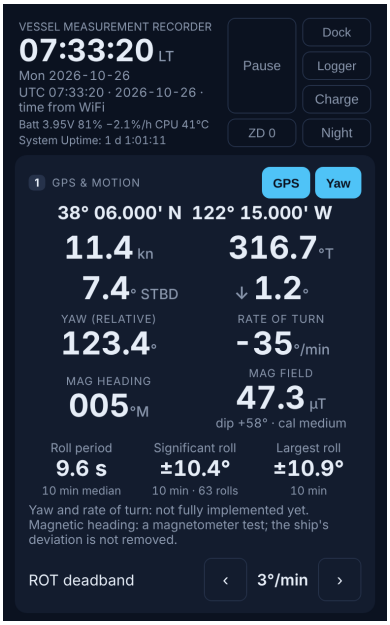
Hiding a card only hides it: recording, marks, the OLED and the LEDs carry on.

Day / Night / Red Mode

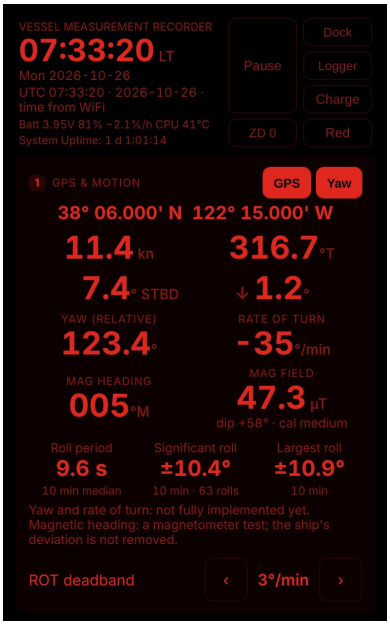
Changes the colors for everything – even the generated plots auto-change based upon the setting at the top



Day



Night

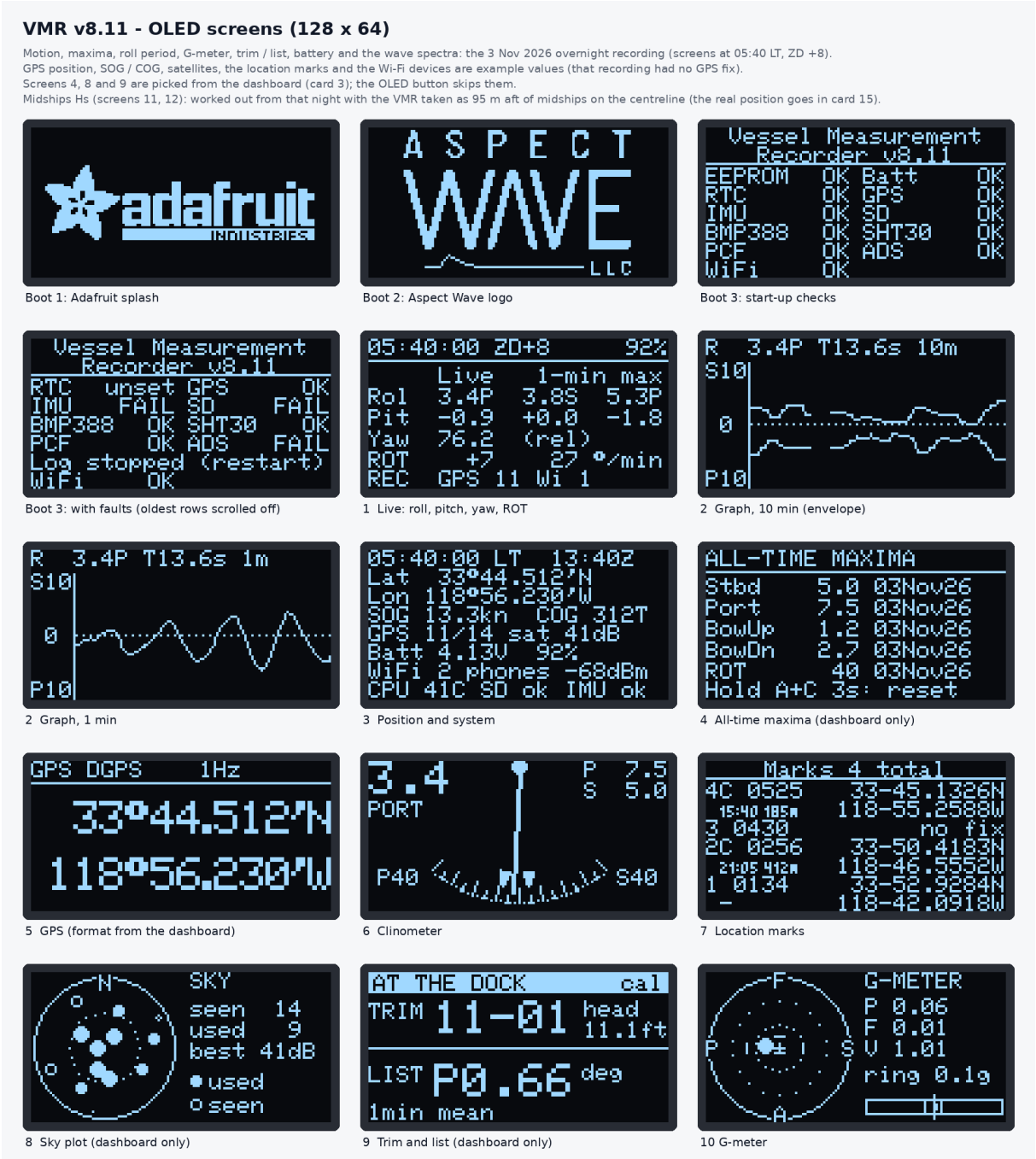


Red

OLED Screens

The possible screens that can be displayed on the OLED

Top three are during boot



Boot 1: Adafruit splash

Boot 2: Aspect Wave logo

Boot 3: start-up checks

Boot 3: with faults (oldest rows scrolled off)

1 Live: roll, pitch, yaw, ROT

2 Graph, 10 min (envelope)

2 Graph, 1 min

3 Position and system

4 All-time maxima (dashboard only)

5 GPS (format from the dashboard)

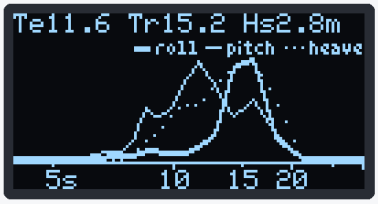
6 Clinometer

7 Location marks

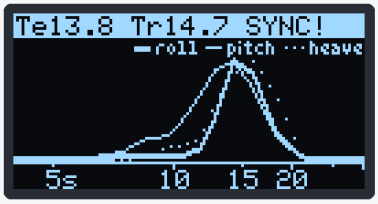
8 Sky plot (dashboard only)

9 Trim and list (dashboard only)

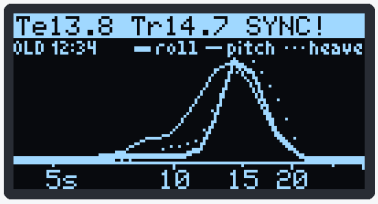
10 G-meter



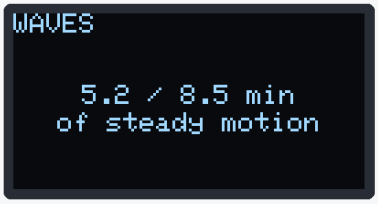
11 Waves (10:29 UTC), Hs at midships



11 Waves: Te near Tr, flagged (09:27 UTC)



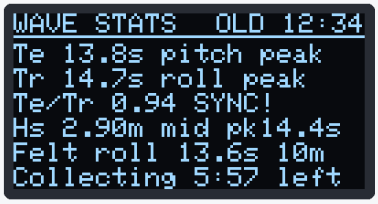
11 Waves: last estimate kept, 12:34 old



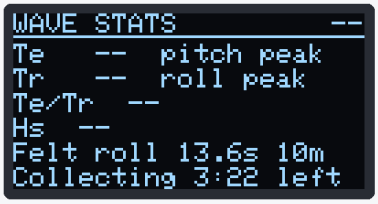
11 Waves: collecting, none yet



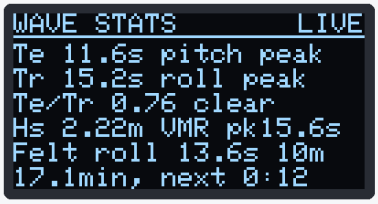
12 Wave stats (10:29 UTC), Hs at midships



12 Wave stats: last estimate, refilling



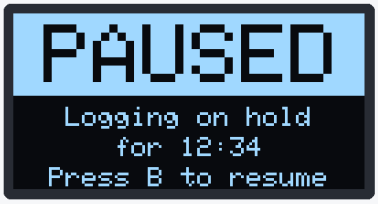
12 Wave stats: collecting, none yet



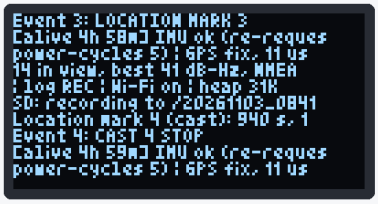
12 Wave stats: VMR position not entered



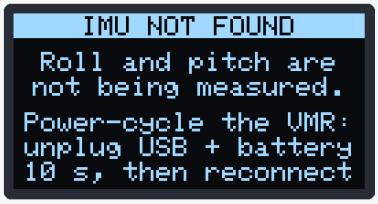
13 Wi-Fi devices (example devices)



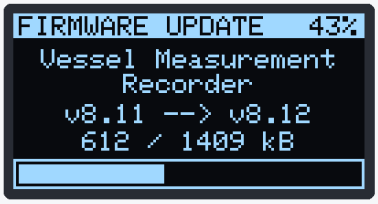
Paused (over every screen)



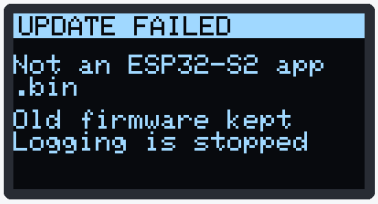
Serial monitor on the OLED



IMU not found (boot)



Firmware update



Firmware update failed



Datalogger: screen button, 5-min rows



Datalogger: screen button, 10 Hz